



K L Deemed to be University

Department of computer science and Engineering -1 -- KLVZA
Course Handout
2025-2026, Term3

Course Title	: MATHEMATICS FOR DATA SCIENCE AND ANALYTICS
Course Code	: 25MT1306E
L-T-P-S Structure	: 2-0-4-0
Pre-requisite	:
Credits	: 4
Course Coordinator	: Venkata Ramana Reddy Gurrampati
Team of Instructors	:
Teaching Associates	:
Syllabus :	CO-1: Introduction to data science; types of data (structured, unstructured); data sources; data lifecycle (collection → cleaning → analysis → interpretation); data cleaning techniques (missing values, duplicates, outliers); data transformation and feature preparation; data quality and integrity; introduction to data analysis workflow. CO--2: Types of data and measurement scales; measures of central tendency (mean, median, mode); measures of dispersion (variance, standard deviation, range, interquartile range); percentiles and quartiles. CO-3: Basic probability concepts; sample space and events; conditional probability; Bayes theorem; discrete and continuous random variables; probability distributions (Bernoulli, Binomial, Normal); expectation and variance; probabilistic reasoning in data analysis. CO-4: Sampling distributions; point estimation; confidence intervals; hypothesis testing (z-test, t-test, chi-square); p-values; Type I & II errors; power of tests (conceptual); interpretation of results. CO-5: Covariance and correlation analysis; simple linear regression; introduction to multiple regression; regression assumptions; residual analysis; multicollinearity concept; model interpretation and evaluation for analytical decision making. CO-6: Advanced visualization; data storytelling; pattern discovery; model interpretation; evaluation metrics (MSE, RMSE, MAE, R ²); reporting insights; decision making; ethics in data analytics.
Text Books :	1. Probability and Statistics for Engineering and the Sciences — Jay L. Devore, Cengage Learning, 2015. 2. Practical Statistics for Data Scientists — Peter Bruce, Andrew Bruce, Peter Gedeck, O'Reilly Media, 2020.
Reference Books :	1. Statistics (4th Edition) — David Freedman, Robert Pisani, Roger Purves, W.W. Norton & Company, 2018. 2. Think Stats (2nd Edition) — Allen B. Downey, O'Reilly Media, 2014. 3. Regression Analysis by Example (5th Edition) — Samprit Chatterjee, Ali S. Hadi, Wiley, 2015. 4. Data Science from Scratch (2nd Edition) — Joel Grus, O'Reilly Media, 2019.
Web Links :	1. https://www.edx.org/learn/probability?utm_source=chatgpt.com 2. https://pll.harvard.edu/subject/statistics-probability?utm_source=chatgpt.com 3. https://pll.harvard.edu/course/data-analysis-basic-probability-and-statistics?utm_source=chatgpt.com 4. https://www.mygreatlearning.com/academy/learn-for-free/courses/probability-for-data-science?utm_source=chatgpt.com 5. https://harvardonline.harvard.edu/course/data-science-probability?utm_source=chatgpt.com 6. https://pll.harvard.edu/course/introduction-probability?utm_source=chatgpt.com
MOOCS :	https://www.coursera.org/learn/machine-learning-probability-and-statistics?utm_source= https://www.coursera.org/learn/statistics-for-data-science-python?utm_source= https://www.coursera.org/specializations/advanced-statistics-data-science?utm_source=
Course Rationale :	This course provides a foundation in data science, statistics, and probability for data-driven decision making. It covers the data lifecycle, statistical

	analysis, probability concepts, and regression techniques. The course also emphasizes data visualization, interpretation, and ethical use of data, enabling learners to analyze, interpret, and communicate insights effectively.
Course Objectives :	This course aims to develop a strong foundation in data science, statistics, and probability by enabling learners to understand data types, sources, and lifecycle, apply data cleaning and transformation techniques, perform statistical and probabilistic analysis, use inferential methods such as hypothesis testing and confidence intervals, build and interpret regression models, and effectively communicate insights through visualization and data storytelling while adhering to ethical principles in data analytics.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Apply data science foundations including data preprocessing, cleaning, transformation, and feature preparation techniques to prepare real-world datasets for analysis.	PO2	3
CO2	Analyse and interpret descriptive statistical measures and perform exploratory data analysis to summarize and understand patterns in datasets..	PO2	4
CO3	Analyse probability theory and random variable models to quantify uncertainty and examine probabilistic events in data-driven problems.	PO2	4
CO4	Analyse statistical inference methods, including parameter estimation, confidence intervals, and hypothesis testing, to draw conclusions from sample data.	PO2	4
CO5	Evaluate relationships between variables using covariance, correlation, and regression techniques for data analysis and prediction.	PO5	5
CO6	Evaluate data analytics, visualization, and interpretation techniques to communicate insights, assess models, and support data-driven decision-making	PO5	5

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
CO1	3	Btl-2 Describe fundamentals of data science, data types, sources, lifecycle, and data quality.	Btl-3 Apply data cleaning, transformation, and feature preparation techniques.		
CO2	4	Btl-2 Describe types of data, measurement scales, and statistical measures such as central tendency and dispersion.	Btl-3 Apply statistical techniques to compute mean, median, mode, variance, standard deviation, range, and interquartile range for datasets.	Btl-4 Analyze datasets using percentiles and quartiles to understand data distribution and variability.	

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
CO3	4	Btl-2 Illustrate the probability concepts, random variables, and probability distributions.	Btl-3 Apply probability laws to compute expectation, variance, and solve problems.	Btl-4 Analyze uncertainty and support decision-making using probabilistic reasoning.	
CO4	4	Btl-2 Outline sampling methods, sampling distributions, Central Limit Theorem, and concepts of estimation and hypothesis testing.	Btl-3 Apply confidence intervals and p-values to test statistics for making decisions under uncertainty.	Btl-4 Evaluate statistical results by interpreting significance levels, errors, and drawing conclusions for data-driven decisions.	
CO5	5	Btl-2 Summarize covariance, correlation, regression concepts, and their assumptions.	Btl-3 Implement simple and multiple regression models and perform residual analysis.	Btl-4 Assess regression models by analyzing multicollinearity and validating assumptions.	Btl-5 Design appropriate regression models and justify their suitability for analytical decision-making.
CO6	5	Btl-2 Outline advanced visualization techniques, data storytelling concepts, evaluation metrics, and ethical considerations in data analytics.	Btl-3 Apply visualization methods and evaluation metrics (MSE, RMSE, MAE, R^2) to analyze and present data effectively.	Btl-5 Examine patterns in data, interpret model outputs, and derive meaningful insights for analytical reporting.	Btl-5 Develop data-driven reports and justify decisions by integrating visualization, model interpretation, and ethical considerations.

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Introduction to course handout and Concepts of data, types, and quality dimensions.	Text book[1], Ch 1, pp. 9–10	Chalk, Talk	End Semester Exam, SEM-EXAM1
2	CO2	COI-2	Data Quality Metrics	Ch 1, pp. 15–30	Chalk, Talk	End Semester Exam, SEM-EXAM1

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
3	CO3	COI-3	Data Preprocessing	Textbook[1], Ch 1, pp. 30–40	Chalk,Talk	ALM,End Semester Exam,LCE,SEM-EXAM1
4	CO1	COI-3	Data Cleaning Methods.	Textbook[1], Ch 1, pp. 35–45	Chalk,Talk	End Semester Exam,SEM-EXAM1
5	CO1	COI-3	Data Transformation.	Textbook[1], Ch 1, pp. 40–50	Chalk,Talk	End Semester Exam,LCE,SEM-EXAM1
6	CO1	COI-3	Feature Preparation.	Textbook [1], Ch 1, pp. 45–55	Chalk,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
7	CO2	COI-2	Measures of Central Tendency.	Textbook[1], Ch 1, pp. 55–65	Chalk,Talk	End Semester Exam,SEM-EXAM1
8	CO2	COI-3	Measures of Dispersion.	Textbook [1], Ch 1, pp. 65–75	Chalk,Talk	End Semester Exam,SEM-EXAM1
9	CO2	COI-4	Discuss about analyze the central tendency and variability	Textbook[1], Ch 1, pp. 75–85	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
10	CO2	COI-3	Introduction to Correlation between two variables	Textbook[1], Ch 12, pp. 520–530	Talk	End Semester Exam,LCE,SEM-EXAM1
11	CO2	COI-4	Covariance & Correlation	Textbook[1], Ch 5, pp. 250–260	Chalk,Talk	End Semester Exam,Home Assignment,SEM-EXAM1
12	CO2	COI-4	Distribution Analysis	Textbook[1], Ch 1, pp. 25–40	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
13	CO3	COI-2	Introduction to basic concept of probability and related properties.	Textbook [1], Ch 2, pp. 90–120	Chalk,Talk	End Semester Exam,SEM-EXAM2
14	CO3	COI-3	Conditional Probability, Bayes's theorem and related applications.	Textbook[1], Ch 2, pp. 120–140	Chalk,Talk	End Semester Exam,SEM-EXAM2
15	CO3	COI-3	Discrete and continuous random variables	Textbook[1], Ch 3 & 4, pp. 150–210	Chalk,Talk	ALM,End Semester Exam,LCE,SEM-EXAM2
16	CO3	COI-3	Expectation & Variance of	Textbook[1],	Chalk,Talk	End Semester Exam,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			random variables of the distribution	Ch 3, pp. 160–180		
17	CO3	COI-4	Standard Distributions of random variables	Textbook[1], Ch 3 & 4, pp. 180–220	Chalk,Talk	End Semester Exam,Home Assignment,SEM-EXAM2
18	CO3	COI-4	Normal distribution with related case study applications.	Textbook [1], Ch 4, pp. 200–220	Chalk,Talk	End Semester Exam,SEM-EXAM2
19	CO4	COI-2	Introduction to sampling distribution	Textbook[1], Ch 5, pp. 230–250	Chalk,Talk	End Semester Exam,SEM-EXAM2
20	CO4	COI-3	Sampling Distributions on small samples (t-Test), chi-square test.	Textbook[1], Ch 5, pp. 250–270	Chalk,Talk	End Semester Exam,LCE,SEM-EXAM2
21	CO4	COI-3	Sampling Distributions on large samples (Z-test)	Textbook[1], Ch 5, pp. 250–270	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM2
22	CO4	COI-3	Introduction to Hypothesis Testing	Textbook [1], Ch 8, pp. 350–380	Chalk,Talk	End Semester Exam,SEM-EXAM2
23	CO4	COI-4	Confidence Intervals & p-values	Textbook[1], Ch 7, pp. 310–340	Chalk,Talk	End Semester Exam,SEM-EXAM2
24	CO4	COI-4	Analysis of Variance	Textbook[1], Ch 10 & 11, pp. 440–500	Chalk,Talk	ALM,End Semester Exam,LCE,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 2 Concepts of data, types, and quality dimensions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Concepts of data, types, and quality dimensions	2	Chalk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Problem solving	2	Talk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 2 Discuss about the Applications of quality and dimensions

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
20	Discuss about the Applications of quality and dimensions	2	Chalk	--- NOT APPLICABLE ---
20	Problem solving	2	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 3 Data Preprocessing

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the missing data	3	Chalk	--- NOT APPLICABLE ---
30	Discuss about the concept of Identification of missing values and errors	3	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 3 Data Cleaning Methods.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the missing values	3	Chalk	--- NOT APPLICABLE ---
30	Mathematical handling of missing and inconsistent data	3	Talk	Quiz/Test Questions
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 3 Data Transformation.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the basics of scaling	3	Chalk	--- NOT APPLICABLE ---
30	Discuss about the Scaling and transformation concepts	3	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 3 Feature Preparation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the Mathematical form of variables	3	Chalk	One minute paper
30	Discuss about concept of Encoding and derived variables (mathematical form)	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 2 Measures of Central Tendency.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Introduction of Measures of central tendency	2	Chalk	--- NOT APPLICABLE ---
30	Discuss about the definitions Mean, median, mode definitions and computation	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 3 Range, variance, standard deviation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the concepts of Range, variance, standard deviation	3	Chalk	--- NOT APPLICABLE ---
30	Problem Solving on Range, variance, standard deviation.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 3 Discuss about analyze the central tendency and variability

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the Case study and applications of central tendency and variability.	4	Chalk	Quiz/Test Questions
30	Problem solving on central tendency and variability.	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

SESSION NUMBER : 10

Session Outcome: 3 Apply the concept of Correlation between two variables

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Introduction to Correlation between two variables	2	Chalk	--- NOT APPLICABLE ---
30	Problem solving on correlation between two variables	3	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 4 Discuss and analyze the concept of covariance and correlation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Explain the concept of covariance and correlation.	3	Chalk	--- NOT APPLICABLE ---
30	Problem solving on covariance and correlation.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 4 Analyze the Distribution Analysis of frequency of the given data

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Analyze the Frequency distribution and shape analysis	4	Chalk	--- NOT APPLICABLE ---
30	Problem solving on frequency distribution	4	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 2 Introduction to basic concept of probability and related properties.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Introduction to basic concept of probability and related properties.	2	Chalk	--- NOT APPLICABLE ---
30	Problem solving on probability and its applications	2	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 3 Conditional Probability , Bayes' theorem and related applications.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Conditional Probability and related applications.	3	Chalk	--- NOT APPLICABLE ---
30	Problem solving on conditional probability.	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 2 Discrete and continuous random variables

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Explain the concept of discrete and continuous random variables	2	Chalk	--- NOT APPLICABLE ---
30	Problem solving on discrete and continuous random variables	2	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 3 Expectation & Variance of random variables of the distribution

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Expectation & Variance of random variables of the distribution	3	Chalk	--- NOT APPLICABLE ---
30	Problem solving on Expectation & Variance of random variables of the distribution	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 4 Standard Distributions with case study problems

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Explain the concept of Standard Distributions with case study problems	4	Chalk	--- NOT APPLICABLE ---
30	Problem solving on Standard Distributions with case study problems	4	Chalk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 4 Normal distribution with related case study applications.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Normal distribution with related case study applications.	4	Chalk	--- NOT APPLICABLE ---
30	Problem solving on standard normal distribution	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 2 Sampling methods and sample statistics

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Discuss about the concept of Sampling methods and sample statistics	2	Chalk	--- NOT APPLICABLE ---
30	Problem solving on Sampling methods and sample statistics	2	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 3 Sampling Distributions on small samples

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Sampling Distributions on small samples	3	Chalk	--- NOT APPLICABLE ---
30	Problems solving on Sampling distributions on small samples	3	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 3 Sampling Distributions on large samples

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Sampling Distributions on large samples	3	Chalk	Case Study
30	Problem solving on Sampling Distributions on large samples	3	Chalk	Case Study
5	Summary of the session	1	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

Session Outcome: 3 Null and alternative hypothesis, test statistics

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Explain and discuss the concept of Null and alternative hypothesis, test statistics	3	Chalk	--- NOT APPLICABLE ---
30	Problem solving on Null and alternative hypothesis, test statistics	3	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 4 Analyze the Interval estimation and decision making

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
10	Analyze the Interval estimation and decision making	4	Chalk	--- NOT APPLICABLE ---
30	Problem solving on the Interval estimation and decision making	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 4 Analyze the concept of one way and two-way Analysis of Variance

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
10	Analyze the concept of one way and two-way Analysis of Variance	4	Chalk	Brain storming session
30	Problem solving on Analyze the concept of one way and two-way Analysis of Variance	4	Chalk	Brain storming session
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Data Cleaning and Preprocessing	CO5
2	Feature Engineering and Transformation	CO5

Tutorial Session no	Topics	CO-Mapping
3	Statistical Summary Computation	CO5
4	Exploratory Data Analysis & Visualization	CO5
5	Visualization Techniques	CO5
6	Probability computation	CO5
7	Conditional Probability and related properties	CO5
8	Simulation of Random Variables	CO5
9	Sampling and CLT Experiment	CO5
10	Hypothesis Testing	CO5
11	Online Sales Conversion Rate	CO5
12	Correlation Analysis	CO5
13	Linear Regression Modeling	CO6
14	Data Visualization and Storytelling	CO6
15	Advanced Visualizations—Heatmaps & Pair Plots	CO6
16	Regression Evaluation Metrics	CO6
17	Pattern & Anomaly Detection	CO6
18	Data Storytelling Report	CO6
19	Dashboard Development	CO6
20	End-to-End Analytics Pipeline	CO6

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 5 Load a raw dataset with missing values, duplicates, and inconsistencies. Apply techniques to clean and prepare data for analysis.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

45	Data Cleaning – Basics	4	Chalk	--- NOT APPLICABLE ---
45	Data Cleaning – Advanced	4	PPT	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 5 Transform variables, normalize numerical features, and encode categorical variables for analysis readiness.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Feature Engineering – Basics	4	PPT	--- NOT APPLICABLE ---
45	Feature Engineering – Encoding	4	Chalk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 5 Compute mean, median, variance, standard deviation, and summarize dataset characteristics.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
45	Statistical Summary – Mean/Median	4	Chalk	--- NOT APPLICABLE ---
45	Statistical Summary – Variability	4	Talk	--- NOT APPLICABLE ---
5	Summery of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 5 Perform EDA using histograms, boxplots, scatter plots to identify trends and outliers.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Boxplots, scatter plots	4	Chalk	--- NOT APPLICABLE ---
45	Histograms, bar charts	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 5 Apply advanced visualization techniques such as histograms, boxplots, and scatter plots to analyze distributions, detect outliers, and uncover relationships between variables.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Create histograms and boxplots	4	Chalk	--- NOT APPLICABLE ---
45	scatter plots to detect patterns/outliers	4	PPT	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 5 Compute and interpret probabilities of events from real datasets using fundamental probability rules to model uncertainty in data-driven contexts.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Calculate probabilities of events using dataset (e.g., customer purchase probability).	4	Chalk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
45	Compute simple conditional probabilities from dataset events	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 5 Analyze conditional probabilities and apply Bayes' theorem to evaluate dependent events and update probabilistic reasoning based on new information.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Compute conditional probabilities such as likelihood of purchase given demographic attributes.	4	Chalk	--- NOT APPLICABLE ---
45	Apply Bayes' theorem to update probabilities based on new information or evidence.	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 5 Simulate discrete random variables such as Bernoulli and Binomial distributions to compare empirical outcomes with theoretical probability models.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Simulate Bernoulli trials to model binary outcomes such as success/failure experiments.	4	Chalk	--- NOT APPLICABLE ---
45	Apply Bayes' theorem to update probabilities based on new information or evidence.	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 5 Implement various sampling techniques to study sample behavior and evaluate their impact on statistical estimation and inference

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Evaluate their impact on statistical estimation and inference	4	Chalk	--- NOT APPLICABLE ---
45	Apply and analyze to Sample means distribution	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 5 Analyze data using statistical methods such as t-tests by interpreting p-values and confidence intervals to support data-driven decision-making

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
45	Null and alternative hypotheses and understanding significance levels and decision criteria	4	Chalk	--- NOT APPLICABLE ---
45	Analyze data using statistical methods such as t-tests by interpreting p-values and confidence intervals to support data-driven decision-making	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 5 An e-commerce company claims that the conversion rate is 20%.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Null Hypothesis (H_0): Conversion rate = 20% and Alternative Hypothesis (H_1): Conversion rate $\neq$ 20%	4	Chalk	--- NOT APPLICABLE ---
45	Collect sample data of website visitors and purchases and Compute test statistic (z-test for proportion)	4	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 5 Compute and interpret covariance to analyze the directional relationship between variables in a dataset.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Compute and interpret covariance to analyze the directional relationship between variables in a dataset.	4	Chalk	--- NOT APPLICABLE ---
45	Evaluate the strength and direction of relationships using correlation coefficients and visualize them through correlation matrices	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 5 Develop and apply simple linear regression models to establish relationships between dependent and independent variables using mathematical formulations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Apply simple linear regression models to establish relationships between dependent and independent variables using mathematical formulations	4	Chalk	--- NOT APPLICABLE ---
45	Interpret regression results through residual analysis and evaluation metrics such as MSE and R^2 to assess model performance	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 5 Design advanced multi-variable visualizations to identify complex patterns and enhance analytical insights.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Design advanced multi-variable visualizations to identify complex patterns and enhance analytical insights.	4	Chalk	--- NOT APPLICABLE ---
45	Construct compelling data-driven narratives by integrating visualizations with clear and impactful interpretations	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 5 Create advanced visualizations to analyze patterns.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	: Introduction to correlation & multivariate visualization and generate heatmaps (correlation matrix)	4	Chalk	--- NOT APPLICABLE ---
45	Construct compelling data-driven narratives by integrating visualizations with clear and impactful interpretations	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 5 Interpret regression results and evaluate model performance

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Regression evaluation metrics and compute MSE, RMSE, MAE	4	Chalk	--- NOT APPLICABLE ---
45	Calculate and interpret R^2 score and compare model performance and discuss accuracy	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 5 Identify patterns and anomalies using visualization.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Concept of patterns & anomalies and use boxplots, scatter plots, histograms	4	Chalk	--- NOT APPLICABLE ---
45	Detect outliers and unusual trends and interpret anomalies and possible causes	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 5 Prepare a data storytelling report.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
45	Principles of storytelling with data and combine visualizations with statistical insights	4	Talk	--- NOT APPLICABLE ---
45	Structure report (intro, analysis, conclusion) and draft narrative with key findings	4	Chalk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 5 Develop a dashboard-style summary for decision-making.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
45	Introduction to dashboards and create multiple plots (bar, line, heatmap)	4	Chalk	--- NOT APPLICABLE ---
45	Arrange visuals into dashboard layout and Interpret dashboard insights	4	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 5 Perform complete analytics pipeline and present insights.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Chalk	--- NOT APPLICABLE ---
45	Define problem & dataset and data preprocessing & analysis	4	Chalk	--- NOT APPLICABLE ---
45	Visualization & modeling and present insights and recommendations	4	Talk	--- NOT APPLICABLE ---
5	Summary of the session	1	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan:

NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
6	Weekly Homework Assignments	1	Data Science Foundations & Data Preprocessing	Explain data science, types of data, data sources, and the data lifecycle. Discuss data cleaning, transformation, data quality, and the basic data analysis workflow	CO1
12	Weekly Homework Assignments	2	Descriptive Statistics & Exploratory Data Analysis	Explain types of data and measurement scales. Discuss measures of central tendency and dispersion, including mean, median, mode, variance, standard deviation, range, and	CO2

Week	Assignment Type	Assignment No	Topic	Details	co
				interquartile range. Also explain percentiles and quartiles.	
18	Weekly Homework Assignments	3	Probability and Random Variables	Explain basic probability concepts, sample space, and events. Discuss conditional probability and Bayes theorem. Describe discrete and continuous random variables and common probability distributions (Bernoulli, Binomial, Normal). Also explain expectation, variance, and the role of probabilistic reasoning in data analysis.	CO3
24	Weekly Homework Assignments	4	Statistical Inference	Sampling and sampling distributions; Central Limit Theorem; parameter estimation; confidence intervals; hypothesis testing; p-values and significance levels; Type I and Type II errors; interpretation of statistical results in data-driven decision making	CO4

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	---	---	--	--	--	--	--	--	--
	Tutorial	---	---	--	--	--	--	--	--	--
	Lab	---	---	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	--	--	--	--	--
	Skilling	---	---	--	--	--	--	--	--	--
Tue	Theory	---	---	--	--	--	--	--	--	--
	Tutorial	---	---	--	--	--	--	--	--	--
	Lab	---	---	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	--	--	--	--	--
	Skilling	---	---	--	--	--	--	--	--	--
Wed	Theory	V-S1,V-S2,V-S3,V-	V-S1,V-S2,V-S3,V-	---	---	--	--	--	--	--

	Hour	1	2	3	4	5	6	7	8	9
		S4,V-S5	S4,V-S5							
	Tutorial	--	--	---	---	--	--	--	--	--
	Lab	--	--	---	---	--	--	--	--	--
	Skilling	--	--	---	---	--	--	--	--	--
Thu	Theory	V-S6,V-S7,V-S8,V-S9	V-S6,V-S7,V-S8,V-S9	---	---	--	--	--	--	--
	Tutorial	--	--	---	---	--	--	--	--	--
	Lab	--	--	---	---	--	--	--	--	--
	Skilling	--	--	---	---	--	--	--	--	--
Fri	Theory	---	---	--	--	--	--	--	--	--
	Tutorial	---	---	--	--	--	--	--	--	--
	Lab	---	---	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9	--	--	--	--	--
	Skilling	---	---	--	--	--	--	--	--	--
Sat	Theory	---	---	--	--	--	--	--	--	--
	Tutorial	---	---	--	--	--	--	--	--	--
	Lab	---	---	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9	--	--	--	--	--
	Skilling	---	---	--	--	--	--	--	--	--
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	End Semester Exam	Weightage	24		180	6	6	6	6		
		Max Marks	100			25	25	25	25		
	SEM End Project	Weightage	16		90					8	8
		Max Marks	50							25	25
In Semester Formative Evaluation Total= 24 %	Continuous Evaluation - Lab Exercise	Weightage	8		50	2	2	2	2		
		Max Marks	40			10	10	10	10		
	Home Assignment and Textbook	Weightage	8		180	2	2	2	2		
		Max Marks	40			10	10	10	10		
	ALM	Weightage	8		30	2	2	2	2		
		Max Marks	40			10	10	10	10		
In Semester Summative Evaluation Total= 36 %	Lab In Semester Exam	Weightage	8		90					4	4
		Max Marks	50							25	25
	Semester in Exam-II	Weightage	14		90			7	7		
		Max Marks	50					25	25		
	Semester in Exam-I	Weightage	14		90	7	7				
		Max Marks	50			25	25				

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Srinivasa Varanasi	L	7-MA	-	-	-	-
Srinivasa Varanasi	P	7-B,7-A	-	-	-	-
Venkata Ramana Reddy Gurrampati	L	1-MA	-	-	-	-
Venkata Ramana Reddy Gurrampati	P	1-B,1-A	-	-	-	-
Sridhar Wuriti	L	3-MA	-	-	-	-
Sridhar Wuriti	P	3-B,3-A	-	-	-	-
Nerella Srimannarayana	L	2-MA	-	-	-	-
Nerella Srimannarayana	P	2-A,2-B	-	-	-	-
Ankita Tiwari	L	4-MA	-	-	-	-
Ankita Tiwari	P	4-A,4-B	-	-	-	-
Imran Ali	L	6-MA	-	-	-	-
Imran Ali	P	6-A,6-B	-	-	-	-
V Viswanath Sheno	L	8-MA	-	-	-	-
V Viswanath Sheno	P	8-B,8-A	-	-	-	-
SEELAM DEVI	L	5-MA	-	-	-	-
SEELAM DEVI	P	5-B,5-A	-	-	-	-
SUHASINI ALLADA	L	9-MA	-	-	-	-
SUHASINI ALLADA	P	9-A,9-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Venkata Ramana Reddy Gurrampati)(3155)(IR&D)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of Maths

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal)

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